

Unit Rates from Fractions and Mixed Numbers

Rates from fractions, rates with mixed numbers, comparing unit rates, and using a unit rate to solve

Directions

- A unit rate is one quantity divided by the other: $\text{rate} = \frac{\text{amount}}{\text{time or size}}$. When both are fractions, dividing means multiplying by the reciprocal.
- Change every mixed number to an improper fraction *before* dividing.
- Leave answers as exact fractions or mixed numbers in lowest terms — do not round.

1. A runner covers $\frac{3}{4}$ of a mile in $\frac{1}{2}$ of an hour. What is the runner's speed in miles per hour?

Answer: _____ mph

2. A cyclist rides $\frac{5}{8}$ of a kilometre in $\frac{1}{4}$ of an hour. What is the cyclist's speed in kilometres per hour?

Answer: _____ km/h

3. A toy car travels $\frac{7}{8}$ of a metre in $\frac{1}{4}$ of a second. What is its speed in metres per second?

Answer: _____ m/s

4. A cart rolls $2\frac{1}{2}$ feet in $\frac{3}{4}$ of a second. What is its speed in feet per second?

Answer: _____ ft/s

5. A printer produces $3\frac{1}{3}$ pages in $\frac{2}{5}$ of a minute. How many pages does it produce per minute?

Answer: _____

6. One runner covers $\frac{7}{8}$ of a mile in $\frac{3}{4}$ of an hour. A second runner covers $\frac{5}{6}$ of a mile in $\frac{2}{3}$ of an hour.

(a) Speed of the first runner:

Answer: _____ mph

(b) Speed of the second runner:

Answer: _____ mph

(c) Write $<$ or $>$ between the first runner's speed and the second runner's speed.

Answer: _____

7. A pump fills $\frac{5}{6}$ of a tank in $\frac{1}{3}$ of an hour, working at a constant rate. How many hours does it take to fill one whole tank?

Answer: _____ hours

8. A car uses $\frac{2}{5}$ of a gallon of fuel to travel $9\frac{1}{3}$ miles. How many miles does it travel per gallon?

Answer: _____

9. One shop sells $\frac{3}{4}$ of a kilogram of cashews for 6 dollars. Another shop sells $\frac{2}{3}$ of a kilogram for 5 dollars.

(a) Price per kilogram at the first shop, in dollars:

Answer: _____

(b) Price per kilogram at the second shop, in dollars:

Answer: _____

(c) Write $<$ or $>$ between the first shop's price and the second shop's price.

Answer: _____

10. A pump moves $\frac{7}{8}$ of a pool in $1\frac{3}{4}$ hours at a constant rate. How many hours does it take to move the whole pool?

Answer: _____ hours

11. A recipe uses $1\frac{1}{2}$ cups of flour to make $\frac{2}{3}$ of a batch. How many cups of flour does one whole batch need?

Answer: _____

12. A cyclist rides $5\frac{1}{4}$ miles in $\frac{3}{4}$ of an hour, then rides a further $2\frac{1}{2}$ miles in $\frac{1}{2}$ of an hour.

(a) Speed on the first leg:

Answer: _____ mph

(b) Speed on the second leg:

Answer: _____ mph

(c) Speed for the whole ride:

Answer: _____ mph

(d) Explain why the whole-ride speed is not the average of your answers to (a) and (b).
